Carl Lochstampfor, Jr.

LinkedIn | GitHub | e-Resume | ODU Portfolio

Actively seeking an internship in IT / cybersecurity.

Open to AI security, ML engineering, and applied research roles full-time upon graduation.

PROFESSIONAL SUMMARY

Cybersecurity undergraduate at Old Dominion University (**3.9 GPA**) with **CompTIA Security+, Network+, and A+** certifications. AI security researcher with two arXiv preprints on synthetic fraud detection (COVA and COVA-X, both in the peer-review process) and a live dataset platform (ScamLingua.org). Background includes over a decade in high-risk, regulated financial environments (FNMA, FHLMC, FHA, VA), applying the same rigor to incident response, system hardening, and AI governance. Proven record of managing **3× standard workloads** while sustaining **95%+ accuracy**.

TECHNICAL SKILLS & CERTIFICATIONS

- **Certifications:** CompTIA Security+, Network+, & A+ (Completed)
- **Cybersecurity:** Incident Response, Risk Mitigation, Regulatory Compliance, AI Governance, Vulnerability Assessment, System Hardening, Network Defense, Log Analysis, Data Integrity, Policy Enforcement, Python Scripting
- **Tools & Protocols:** Nmap, Wireshark, pfSense, Snort, tcpdump, Rescuezilla, VirtualBox, Bruter, netstat, tasklist, gpedit.msc, Event Viewer, iptables, SSH, RDP, Docker, ZFS, Sanoid/Syncoid
- **AI/ML Frameworks:** XGBoost, scikit-learn, Hugging Face Transformers, PyTorch, DistilBERT, Longformer, Ollama, Qwen 2.5 14B, Open WebUI, RAG
- **Development:** Python (Secure Socket Programming, ML Pipelines), HTML, CSS, JavaScript, SQL, React, C#/.NET, GitHub
- **Platforms:** Windows 10/11, Windows Server, Kali Linux, Ubuntu Linux, Debian Linux, BSD/pfSense, Unraid, Home Lab Environment

APPLIED CYBERSECURITY COURSEWORK

- **Incident Response & Forensics (NIST SP 800-61):** Performed full IR lifecycle after brute-force attack; executed **Nmap** scans, analyzed logs, captured volatile data (**netstat, tasklist**), and applied **GPO** controls to prevent recurrence.
- **Operating System Hardening:** Improved Windows and Linux security baselines by configuring **Group Policy Objects** for access control and auditing failed logins. Hardened Metasploitable and Linux systems using **iptables**.
- **Network Defense & Firewall Security:** Conducted reconnaissance, disabled insecure services (FTP/Telnet), and configured **SSH** for encrypted remote administration using **pfSense**.
- **Intrusion Detection & Packet Capture:** Deployed **Snort IDS** on a Linux sniffer, captured and analyzed PCAP data using **tcpdump/Wireshark**, and identified port scans and brute-force attempts.
- **Disaster Recovery & Data Resilience:** Designed multi-system backup strategy for Windows/Linux/BSD systems; automated encrypted configuration transfers using **WinSCP** for ransomware resilience.

AI SECURITY RESEARCH

Mentored by Dr. Ayan Roy (Christopher Newport University) — CCI Undergraduate Research Program

COVA: Multi-Agent LLM Framework for Elder-Fraud Detection — *arXiv:2604.11752* | *Peer review in progress*

Lead author. Designed and implemented a multi-agent generation pipeline using **Qwen 2.5 14B via Ollama** to produce **3,201 labeled synthetic scam conversations** across 8 elder-fraud categories. Engineered 28-feature extraction with TF-IDF, achieving **72.5% baseline accuracy** with XGBoost. Built v2 label-audit tooling that corrected 1,594 mislabeled conversations.

COVA-X: Scaling Synthetic Fraud Conversations with Transformer Classification — *arXiv:2606.06879* | *Peer review in progress*

Lead author. Scaled dataset to **10,985 conversations** across dual-GPU workstations. Designed three-role virtual-kidnapping architecture reducing artifact rates by **20 percentage points**. Longformer achieved **79.71% accuracy / 0.779 macro-F1**, confirming data-scale hypothesis from COVA.

ScamLingua.org — Self-built distribution platform for COVA research datasets. Designed with trust-focused web engineering principles and CSP hardening.

CYBERSECURITY PROJECTS

Self-Hosted Research NAS & Private AI Server (Unraid, ZFS, Docker) — *Personal Project*

Designed, built, and hardened a research storage server on a mirrored ZFS pool with a **3-2-1 backup** posture: daily snapshots, replication, monthly integrity scrubs, UPS-triggered clean shutdown, a nightly encrypted off-site copy, and an air-gapped drive. Applied **least-privilege** shares and accounts with no inbound internet exposure. Deployed a private, GPU-accelerated **RAG assistant** (Ollama) over ~360 research papers, treating the AI as an untrusted component with a bounded blast radius.

Secure File Sharing System (Python TCP Sockets) — *Milestone Project*

Engineered an encrypted client-server application using Python TCP sockets with secure user authentication and isolated directories, data-integrity checks, and defensive error handling for resilient file operations.

LockBadges (AutoHotkey v2) — Shipped a Windows utility for on-screen lock-key indicators. Privacy-first: reads toggle bits rather than a keyboard hook, so keystrokes are never intercepted; no network access or telemetry.

PROFESSIONAL EXPERIENCE

Cardinal Housing, LLC | Jacksonville, FL | *2012 – 2024 (Closed)* *Property Management and Estate Fiduciary*

Self-employed operator of a three-property rental portfolio, plus probate administration for two estates.

- Held 24-hour on-call responsibility for critical building and tenant failures; selected and supervised contractors against fixed budgets.
- Ensured full adherence to housing regulations and probate laws through meticulous record-keeping, safeguarding sensitive financial documentation and confidential legal data.

BlueHub Capital (SUN) | Boston, MA | *2021 – 2022* *Foreclosure Relief Underwriter*

Underwrote foreclosure-relief mortgages for distressed homeowners; trained and mentored new underwriters.

- Revised underwriting and loan-processing workflows to meet rapidly shifting economic demand.
- Contributed to revisions of Encompass, the loan origination platform, streamlining the application process for staff and applicants.
- Assisted in compliance matters such as bank exams and audits, protecting investor and client interests.

EverBank & TIAA Bank | Jacksonville, FL | *2017 – 2021* *Loss Mitigation Underwriter*

Assessed loss-mitigation files for risk and requirement discrepancies, interfacing legal and business units.

- Maintained a **97% average** on Quality Right Party Contact Monitoring Reports, ensuring compliance with **OCC Consent Orders**.
- Managed a loan portfolio **3× the average workload**, sustaining **95% performance accuracy** through strong process management.
- Operated eight federal and investor underwriting, credit, and court-record systems daily.
- Represented the bank's foreclosure team in mediation and hearings alongside borrowers' counsel and presiding judges; validated court-ordered requests and obtained bankruptcy approvals.

EverBank & Ditech Financial | Jacksonville, FL | *2011 – 2014* *SPOC Relationship Manager*

Sole point of accountability for borrower and third-party foreclosure-prevention inquiries; directed a high-volume loss mitigation pipeline, exceeding investor rate benchmarks by 30%.

EDUCATION

Old Dominion University — B.S. Cybersecurity | GPA: 3.9

Jax Code Academy — Web Development Certificate | 2023

Roanoke College — B.A. Criminal Justice & Philosophy | 2009